

VENUE PARTNER



DS BRAR
CENTER FOR GIRLS
AND WOMEN IN STEM
Supporting the Next Gen of Women in STEM



AI RACE MONTH · GRANDPRIX

PROBLEM STATEMENTS



01 THE SILENT CO-DRIVER

02 WEATHER WHIPLASH

03 CROWD FLOW OPTIMISER

Every build must use Hugging Face. Swipe for the rules →



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Mandatory in every build

MANDATORY FOR ALL TEAMS

GENERAL RULES



01

FRONTEND + BACKEND, BOTH

A frontend the user sees and interacts with, and a backend where the data is processed and the AI does its work. Backend-only or notebook-only submissions will not be accepted.

02

BALANCED DIFFICULTY

Not something solved by calling one ready-made tool. Not a model built completely from scratch. Somewhere in between.

03

HUGGING FACE, PER PERSON AND IN THE BUILD

Every single team member must have their own individual account, and the solution must use something from the Hub – a model, a dataset, a Space, or any suitable tool. No restriction on which.



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IDEAS OVERLAP. EXECUTION DOESN'T.

TIPS TO STAND OUT

Another team will have your idea. That's normal – don't switch direction at the last minute.

FRONTEND

Clean and easy to understand beats clever. It makes a common idea feel polished.

EXECUTION

A small thing that works end to end beats a big thing that half works.

RESEARCH

Understand the real pain point first. It shows in your data choices and your demo.

ITERATION

Don't stop at your first working version. Find what's clunky and fix it.

AND THE ONE THAT DECIDES IT

Clarity of presentation - explain what it does, why it matters and how it works, in simple language.

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01 **PROBLEM STATEMENT 1**

THE SILENT CO-DRIVER

Reading driver stress from radio calls.



THE CHALLENGE

Engineers are watching numbers, not listening to tone. A tired or stressed driver goes unnoticed because nobody has time to hear it.

WHAT YOU BUILD

- Upload or play a driver radio clip.
- Speech converted to readable text.
- Tone analysed - calm, stressed or tired.
- Shown next to lap times, so stress lines up with slower laps.

INPUT

Radio audio clips + lap time data

AI - HUGGING FACE

Speech-to-text and tone analysis

OUTPUT

Transcript, mood label and a lap-performance chart

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02

PROBLEM STATEMENT 2

WEATHER WHIPLASH

Live track condition detector.



THE CHALLENGE

Track conditions flip faster than any weather report. Teams need to know right now whether the track is getting safer or riskier.

WHAT YOU BUILD

- Feed in trackside or onboard image frames.
- Each frame classified: dry, damp, wet or drying.
- A trend over time - better or worse?
- A call to action: "Consider tire change soon."

INPUT

Track photos or video frames,
optional weather data

AI - HUGGING FACE

Image classification per frame

OUTPUT

Condition label, trend graph
and a tire-change suggestion

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PROBLEM STATEMENT 3

CROWD FLOW OPTIMISER

Simulating and rerouting crowds in real time.

THE CHALLENGE

Large venues and events - stadiums, railway stations, festivals - often see people bunch up at entry gates, food counters, or exits without warning. There is no easy way to spot these pile-ups before they become dangerous.

WHAT YOU BUILD

- You feed in the layout of the venue: entry gates, walkways, concession points, and emergency exits.
- The system simulates how crowds move through this layout under different crowd sizes and event schedules.
- It automatically points out where and when bottlenecks are likely to form.
- It suggests real-time rerouting, so people are guided away from crowded areas before things get risky.

INPUT

Venue layout, expected crowd size, and event schedule

AI - HUGGING FACE

Crowd simulation, bottleneck detection and rerouting

OUTPUT

A live map of bottleneck zones and recommended rerouting paths



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PICK YOUR PROBLEM

One statement per team. Bring a working frontend, a real backend and Hugging Face doing genuine work.



ONLINE ROUND

15 AUG

GRANDPRIX · OFFLINE

22 AUG

Winners take direct entry to the TrackShift x Mphasis F1 hackathon in September.

QUESTIONS

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START BUILDING →